

TECHNOLOGY STACK

Date	10 July 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack

The AI Powered Debt Relief & Financial Recovery Platform is developed using a modern, scalable technology stack. It includes frontend and backend technologies, machine learning models, databases, deployment platforms and external APIs. These technologies are selected to ensure high performance, security, scalability and a seamless user experience.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, Tailwind CSS, Redux Toolkit	Provides a responsive and intuitive user interface for managing loans, settlements and communications.
2	Backend / Server-Side	Python (Django REST Framework)	Handles business logic, AI integration, loan analysis, negotiation workflows and API development.
3	Database / Data Storage	PostgreSQL, Redis	Stores user information, loan details, settlement plans and session data with high reliability.
4	Machine Learning / AI Engine	Python (scikit-learn, XGBoost, NLP Models)	Used for repayment capacity prediction, settlement recommendation, risk scoring and document analysis.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
5	Cloud / Hosting / Deployment	AWS (EC2, S3, RDS, Lambda, CloudFront)	Ensures scalable deployment, secure storage, automatic backups and high availability.
6	Authentication & Authorization	JWT, OAuth 2.0	Provides secure login, role-based access control and user session management.
7	APIs & Integrations	OpenAI API, SMS Gateway, Payment Gateway	OpenAI API for NLP and document understanding; SMS for alerts; Payment gateway for settlement payments.
8	File Storage & Document Processing	AWS S3, Tesseract OCR, PyPDF2	Stores uploaded documents securely and extracts text from PDFs/images for analysis.
9	Version Control & CI/CD	Git, GitHub, GitHub Actions	Supports version control, collaboration and automated testing and deployment.
10	Monitoring & Logging	AWS CloudWatch, Sentry	Monitors system performance, logs errors and tracks application health.
11	Communication & Notifications	Email Service (SES), Firebase Cloud Messaging	Sends email notifications, EMI reminders, settlement updates and real-time alerts to users.
12	Analytics & Reporting	Metabase, Power BI	Generates insightful dashboards and business reports for performance tracking and decision making.